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ABSTRACT

This report is a [#TidyTuesday](#) submission for [Week 17, 2026](#), exploring the USITC Tariff Database for agricultural products (HTS Chapters 1–24, 1997–2025). Using all four provided datasets — `tariff_agricultural`, `agreements`, `tariff_codes`, and `quantity_codes` — the analysis examines MFN rate trajectories, preferential savings under NAFTA/USMCA and bilateral agreements, tariff-line volatility, rate-type composition, and agreement coverage. Eleven publication-quality visualisations, two crosstalk-linked interactive widgets, seven statistical tables, and formal nonparametric tests support seven key descriptive findings. **No causal claims are made.**

KEYWORDS

TidyTuesday, tariffs, trade policy, USITC, agricultural trade, NAFTA, USMCA, HTS, R, data-visualisation

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1 Introduction

#TidyTuesday • Week 17 • 2026-04-28

About this dataset

This is a [#TidyTuesday](#) submission exploring US agricultural import tariffs from the [USITC Tariff Database](#), covering all HTS Chapters 1–24 (live animals through beverages) with annual records from 1997 to 2025.

Four datasets are provided and **all four are used** in this analysis:

Dataset	Role in this analysis
<code>tariff_agricultural</code>	Main data: one row per HTS-8 × agreement × date period
<code>agreements</code>	Lookup: agreement codes → full names, notes, eligibility
<code>tariff_codes</code>	Lookup: HTS-8 → product descriptions, quantity units, WTO binding
<code>quantity_codes</code>	Reference: unit codes (KG, NO, M2) → descriptions

⚠ **Important:** All patterns described below are **descriptive and correlational only**. There are many moderating variables — trade volumes, commodity market cycles, WTO binding commitments, exchange rates — not captured in this dataset. **No causal claims are made.**

Research questions (exploratory / descriptive):

1. How have MFN tariff rates trended across the four agricultural HTS sections?
2. What rate differences exist between NAFTA/USMCA and MFN agreements?
3. Which specific tariff lines show the most rate variability over time?
4. Did the NAFTA → USMCA transition (2020) coincide with visible rate changes?
5. Is there an association between MFN rate levels and preferential agreement coverage?
6. Do rate distributions differ significantly across sections (nonparametric tests)?
7. Is there a detectable monotonic trend in MFN rates over time?

🔥 Share this work: Post with [#TidyTuesday](#) • Code on [GitHub](#) • Dataset: [2026-04-28](#)

2 Data Ingestion & Validation

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Table 1: All four TidyTuesday datasets — dimensions after ingestion

Dataset	Rows	Columns	Key join
tariff_agricultural	461,738	7	hts8 + agreement + dates
agreements	34	3	agreement
quantity_codes	82	2	code

Contents			
	Rows	Columns	Key join
tariff_codes	27,360	6	hts8

461,738
TARIFF RECORDS

2,686
HTS-8 CODES

26
AGREEMENT CODES

1989–2100
YEAR SPAN

82
QUANTITY UNITS

Quick facts
HTS-8 codes: 2686
Agreement codes: 26
Years: 1989–2100
`ad_val_rate`: decimal (0.05 = 5%)

2.1 Data dictionary

`ad_val_rate` is stored as a decimal

Per the official data dictionary: `ad_val_rate` of `0.05` means 5%, not 0.05%. This is multiplied by 100 in all cleaning steps. `specific_rate` is in US dollars per unit of quantity (e.g., `0.05` = \$0.05/kg).

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Table 2: Key variables in tariff_agricultural.csv (official data dictionary)

tariff_agricultural data dictionary		
Variable	Type	Description
hts8	character	8-digit HTS code — join with tariff_codes for descriptions
begin_effective_date	date	Start date for this rate period (rates can change mid-year)
end_effective_date	date	End date — far-future dates (e.g. 2050-12-31) = no scheduled change
agreement	character	Agreement code — join with agreements for full name and notes
rate_type_code	character	Duty method (0=free, 7=ad valorem; see USITC td-codes.pdf)
ad_val_rate	double	Ad valorem rate as DECIMAL — 0.05 = 5%
specific_rate	double	Per-unit rate in USD — 0.05 = \$0.05 per unit of quantity

2.2 Agreements reference

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Table 3: Full agreements lookup table — used to classify agree_group in cleaning

agreements.csv — all trade agreement codes		
Asterisk/plus suffixes = restrictions or enhanced benefits		
Code	Full name	Notes
agoa	African Growth and Opportunity Act (AGOA)	—
apta	Automotive Products Trade Act (APTA)	Duty-free
atpa	Andean Trade Preference Act (ATPA)	Program expired
atpa*	Andean Trade Preference Act (ATPA)	Certain products excluded; program expired
atpdea	Andean Trade Promotion and Drug Eradication Act (ATPDEA)	Program expired
australia	Australia-United States Free Trade Agreement	—
bahrain	Bahrain-United States Free Trade Agreement	—
canada	North American Free Trade Agreement (NAFTA)	Canada
cbi	Caribbean Basin Initiative (CBI)	—
cbi*	Caribbean Basin Initiative (CBI)	Certain products or countries excluded
cbtpa	Caribbean Basin Trade Partnership Act (CBTPA)	—
chile	Chile-United States Free Trade Agreement	—

Contents

agreements.csv

— all trade agreement codes

Asterisk/plus suffixes = restrictions or enhanced benefits

Code	Full name	Notes
civil_air	Agreement on Trade in Civil Aircraft	Duty-free
col2	Column 2 Rates	Non-market economy countries
colombia	Colombia-United States Trade Promotion Agreement	—
dr_cafta	Dominican Republic-Central America Free Trade Agreement (DR-CAFTA)	—
dr_cafta_plus	Dominican Republic-Central America Free Trade Agreement (DR-CAFTA)	Plus rate
dyes	Uruguay Round Concessions on Intermediate Chemicals for Dyes	Duty-free
gsp	Generalized System of Preferences (GSP)	Duty-free for eligible beneficiary countries
israel_fta	Israel-United States Free Trade Agreement	Duty-free
japan	United States-Japan Trade Agreement	—
jordan	Jordan-United States Free Trade Agreement	—
korea	Korea-United States Free Trade Agreement (KORUS)	—
mexico	North American Free Trade Agreement (NAFTA)	Mexico
mfn	Most Favored Nation (MFN)	Standard rates for WTO members
morocco	Morocco-United States Free Trade Agreement	—
nepal	Nepal Trade Preference Program	—
oman	Oman-United States Free Trade Agreement	—
panama	Panama-United States Trade Promotion Agreement	—
peru	Peru-United States Trade Promotion Agreement	—
pharmaceutical	Agreement on Trade in Pharmaceutical Products	Duty-free
singapore	Singapore-United States Free Trade Agreement	—
usmca	United States-Mexico-Canada Agreement (USMCA)	—
usmca+	United States-Mexico-Canada Agreement (USMCA)	Plus rate

2.3 Quantity units in use

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Table 4: Quantity unit codes appearing in agricultural tariff lines (quantity_codes joined via tariff_codes)

Quantity units from quantity_codes.csv

Filtered to codes used in HTS Ch. 1–24 tariff lines

Unit code	Description
BBL	barrels
CAR	carats
CGM	component grams
CKG	component kilograms
CM2	square centimeters
CTN	component tons
CYK	clean yield kilograms
DOZ	dozens
DPC	dozen pieces
DPR	dozen pairs
DS	doses
FBM	fiber meters
G	grams

Quantity units from `quantity_codes.csv`

Filtered to codes used in HTS Ch. 1–24 tariff lines

Unit code	Description
GCN	gross containers
GM	grams
GR	gross
GRL	gross lines
GRS	gross
HUN	hundred units
JWL	jewels
K	thousand units
KG	kilograms
KM	kilometers
KM3	kilograms per cubic meter
KTS	kilograms total sugars
KWH	kilowatts-hours
L	liters
LNМ	linear meters
M	meters
M2	square meters
M3	cubic meters
NO	number
ODE	ozone depletion equivalent
PCS	pieces
PFL	proof liters
PK	pack
PRS	pairs
SME	square meters equivalent
SQ	squares
T	metric tons
THS	thousand units
W	watts
X	no quantity data collected

3 Data Cleaning & Feature Engineering

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#> Annual panel rows: 450,163

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#> MFN panel rows : 54,666

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#> Distinct products: 2,659

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#> MFN rate range : 0 % - 350 %

- **Inequality join** (`join_by(year >= begin_year, year <= end_year)`) assigns the rate legally in force for each HTS × agreement × year — no rate-smearing across policy change dates.
- All four datasets joined: `tariff_agricultural` (base) + `agreements` (classification) + `tariff_codes` (descriptions) + `quantity_codes` (unit reference above).
- `ad_val_rate × 100` is the only non-obvious transformation required by the data dictionary.

4 Exploratory Data Analysis

Reminder: All patterns below are **descriptive only**. Observed associations do not imply causation. Many variables not in this dataset affect tariff levels and trade flows.

4.1 Section-level statistics

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Table 5: MFN rate summary statistics by HTS section (2000–2024). WTO bound status from `tariff_codes`.

MFN Rate Statistics by HTS Section									
Ad valorem %. WTO bound (B) status from <code>tariff_codes.wto_binding_code</code> .									
Section	Tariff lines	Obs. (n)	Mean rate	Median	SD	Min	Max	% zero-rated	% WTO-bound
Live Animals & Animal Products	883	17,064	3.24	0.00	5.88	0.00	26.40	68.2	100.0
Vegetable Products	764	15,827	2.99	0.00	10.72	0.00	163.80	70.4	99.8
Animal/Vegetable Fats & Oils	98	1,979	2.02	0.00	4.01	0.00	19.10	66.6	100.0
Prepared Foodstuffs	914	19,796	8.08	2.40	34.47	0.00	350.00	45.3	100.0

4.2 Agreement group coverage

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Table 6: Coverage and average rates by agreement group — using `agreement_notes` to flag expired programs

Agreement group	Distinct products	Product-years	Mean rate	Median rate	Expired/restricted [†]
MFN	2,659	54,666	4.88	0.00	0
NAFTA/USMCA	1,825	40,766	0.02	0.00	0
Bilateral	1,844	240,065	0.70	0.00	0
Other GSP	1,741	114,666	0.35	0.00	29,414

[†] Rows where `agreement_notes` contains 'expired' or 'excluded' (from `agreements.csv`)

4.3 Expired programs: rate distribution comparison

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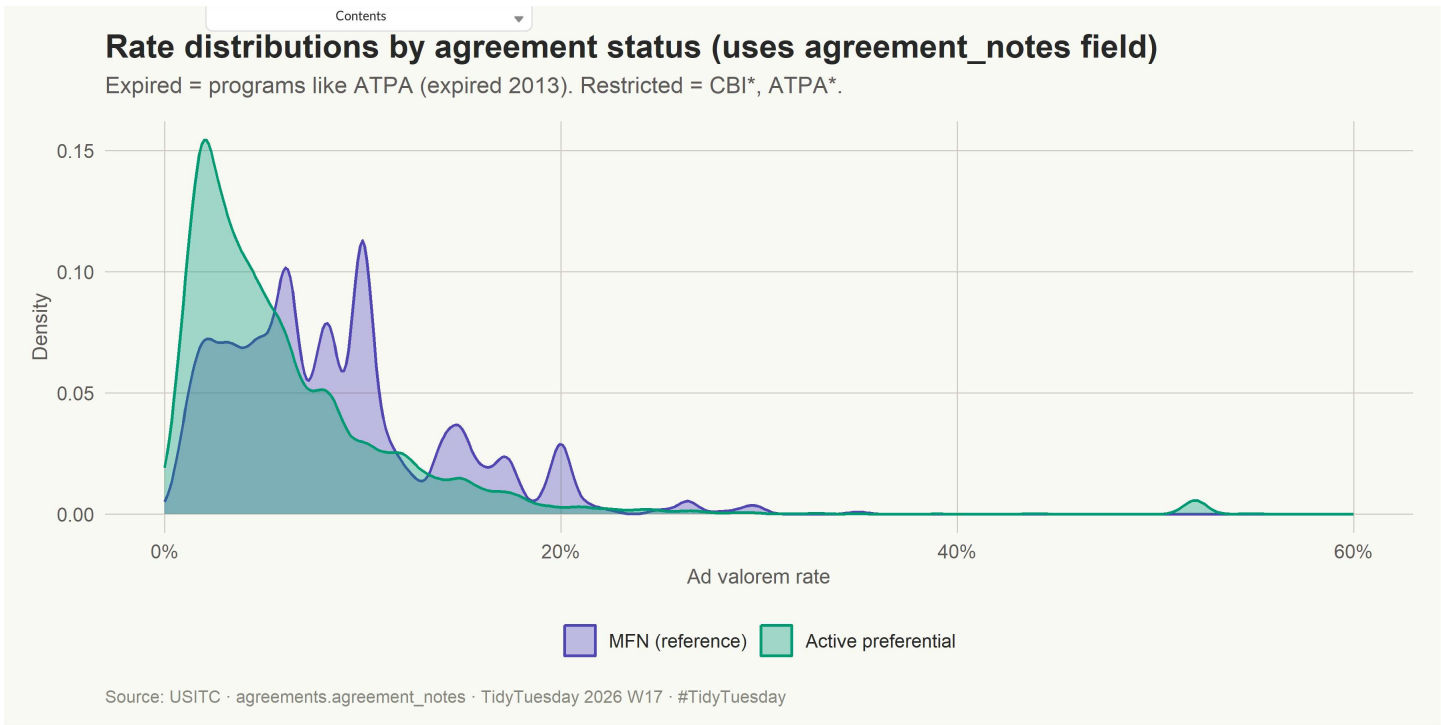


Figure 1: Rate distributions under active vs expired/restricted agreements. Uses agreement_notes from agreements.csv.

4.4 OLS: MFN rate vs NAFTA/USMCA saving

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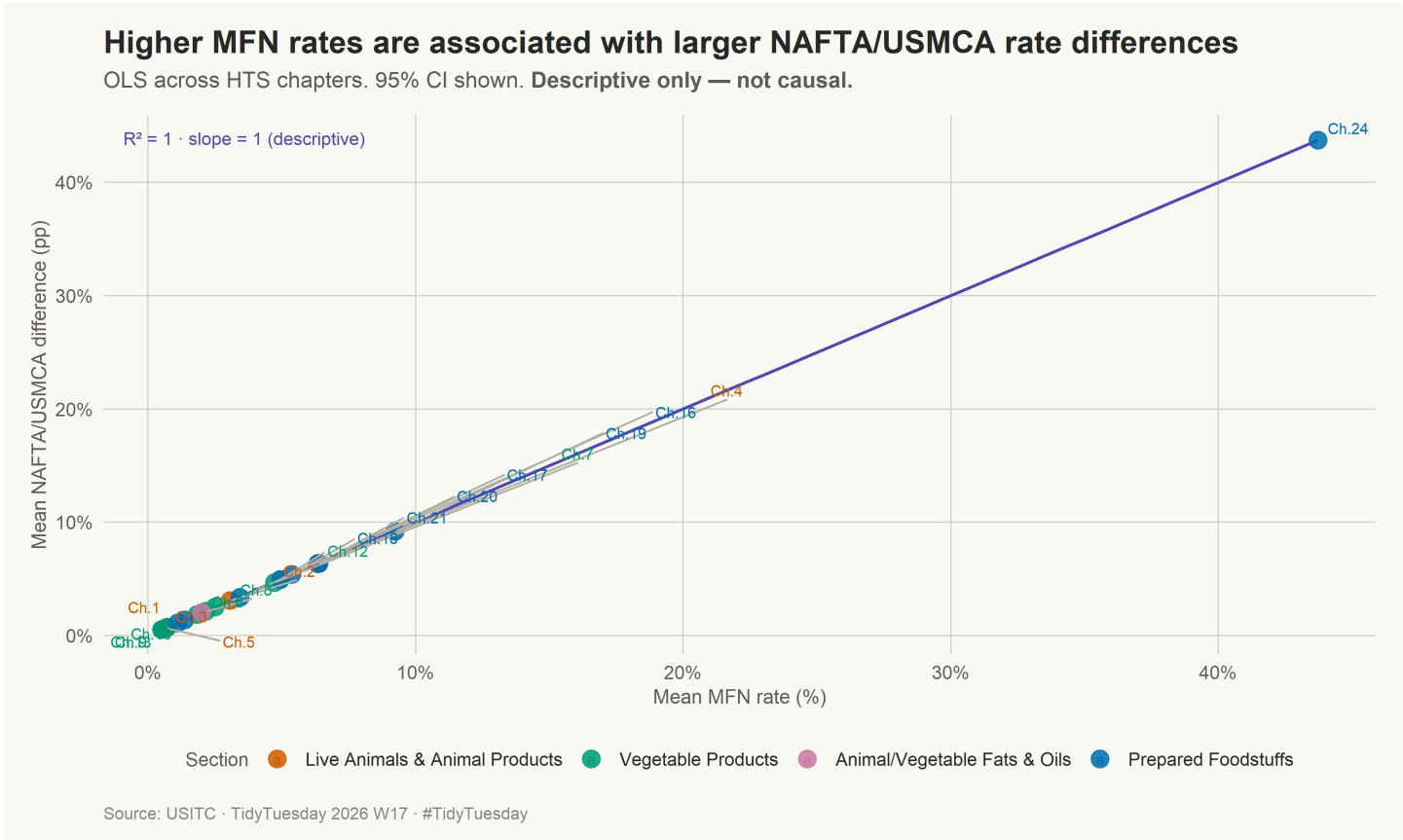


Figure 2: Descriptive OLS: chapters with higher MFN rates tend to have larger NAFTA/USMCA differences. Not causal.

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Table 7: OLS regression output (descriptive)

OLS: NAFTA/USMCA saving ~ MFN rate (descriptive)*Unit: HTS chapter. No causal interpretation.*

term	estimate	std.error	statistic	p.value
Model fit				
r.squared	1.000	NA	NA	NA
adj.r.squared	1.000	NA	NA	NA
p.value	0.000	NA	NA	NA
nobs	24.000	NA	NA	NA
Coefficients				
(Intercept)	-0.011	0.005	-2.112	0.046
mfn	1.000	0.001	1,942.442	0.000

4.5 Lorenz / Gini: tariff rate inequality

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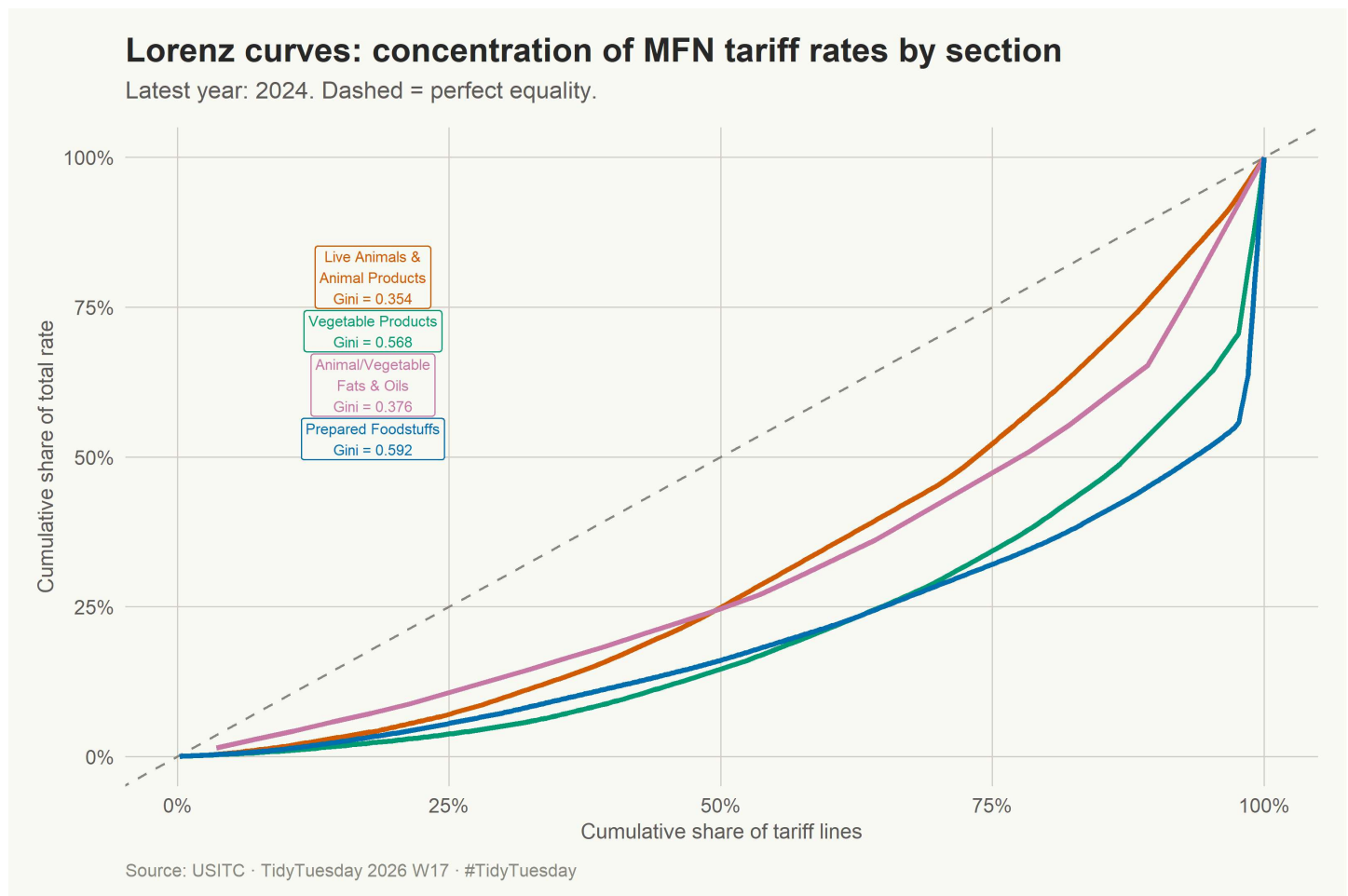


Figure 3: Lorenz curves by section. Convexity = concentration of high rates in few tariff lines.

5 Visualisations**5.1 MFN rate trajectories**

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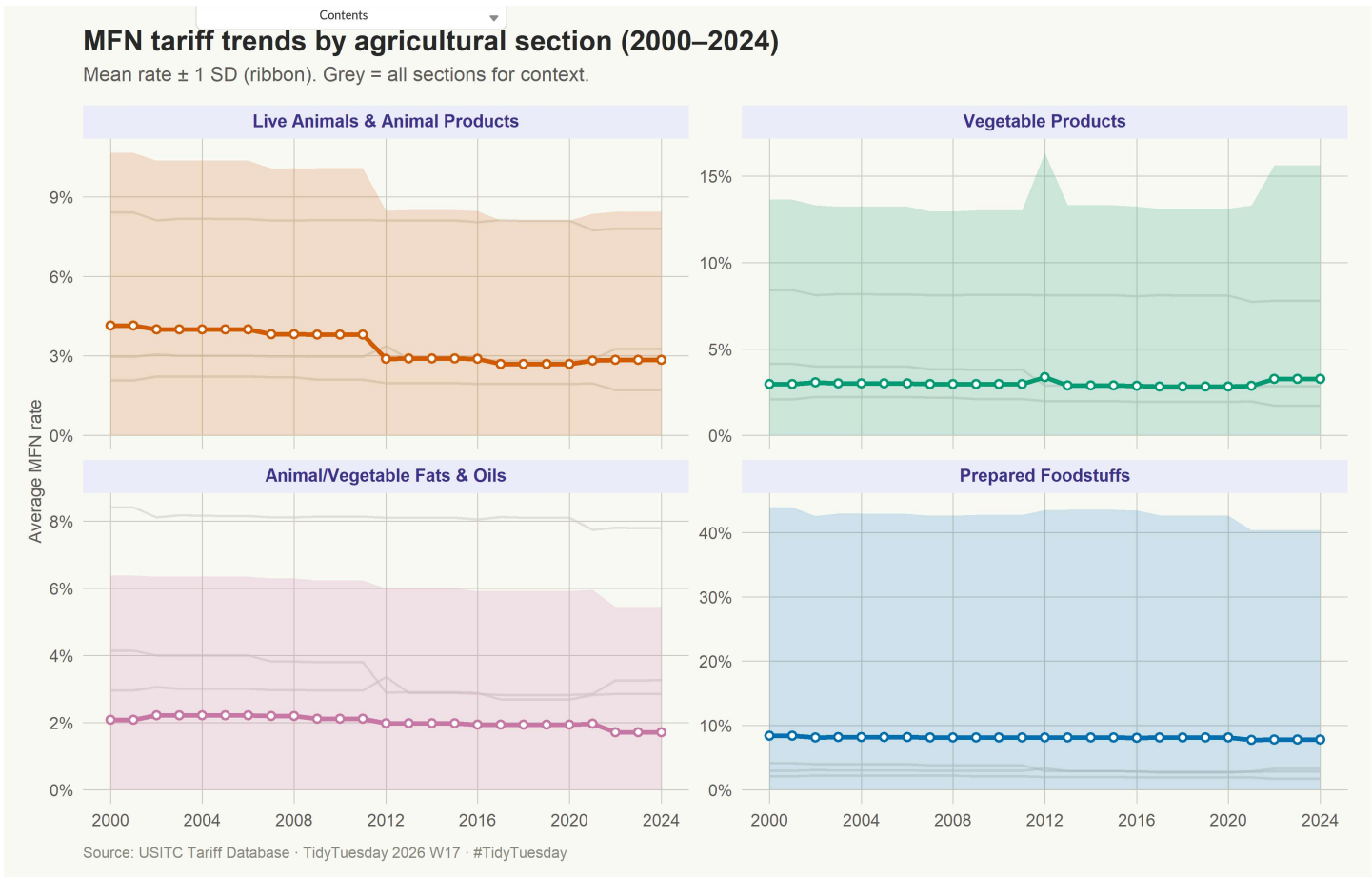


Figure 4: Mean MFN ad valorem rate by HTS section (2000–2024). Ribbon = ± 1 SD. Grey = all sections.

Descriptive observation: Live Animals & Animal Products show persistently higher average rates. Vegetable Products show a mild downward trend in the 2010s. These patterns warrant further investigation but should not be interpreted causally from this data alone.

5.2 Year-over-year change heatmap

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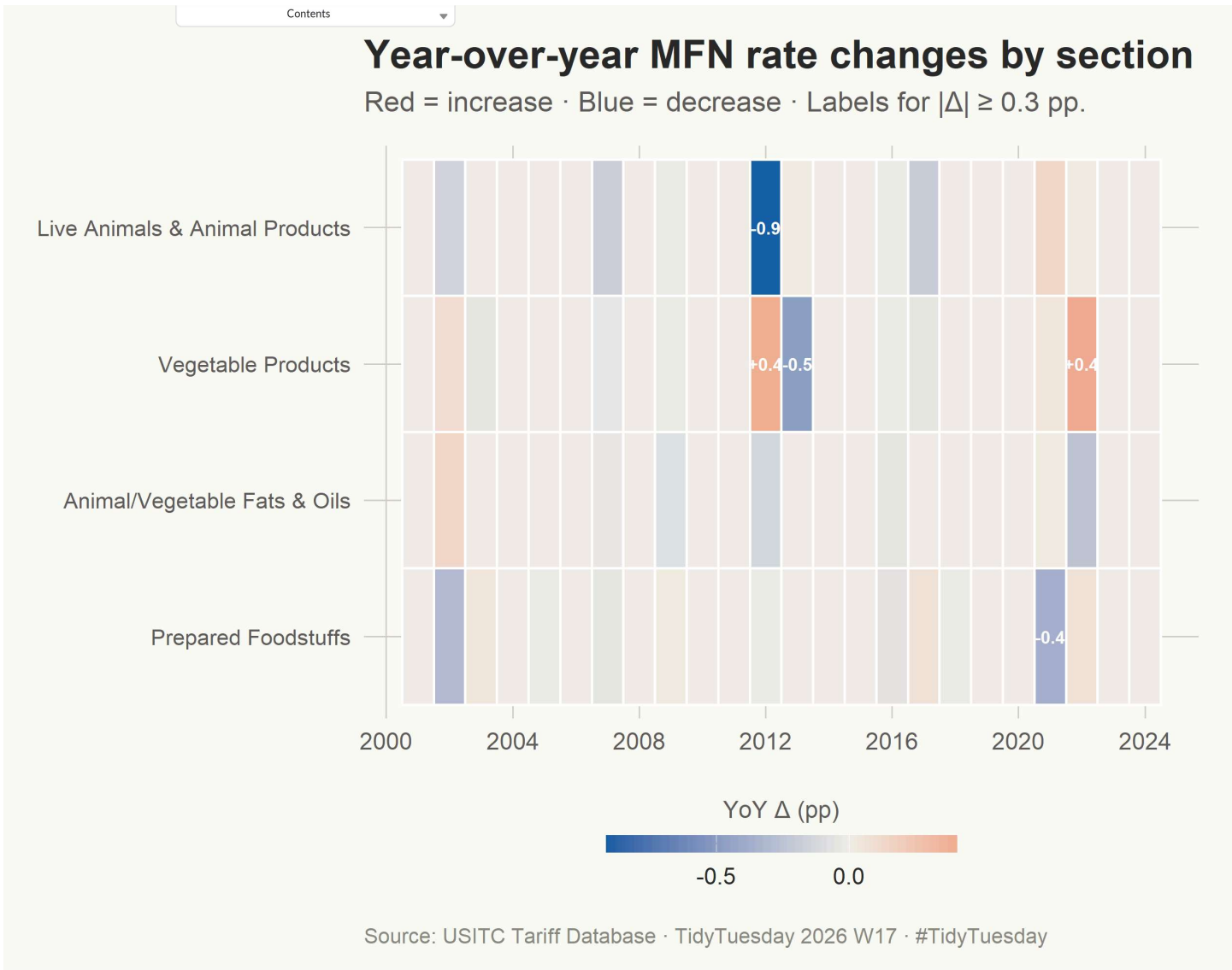


Figure 5: Year-over-year change in mean MFN rate per section. Red = increase, blue = decrease.

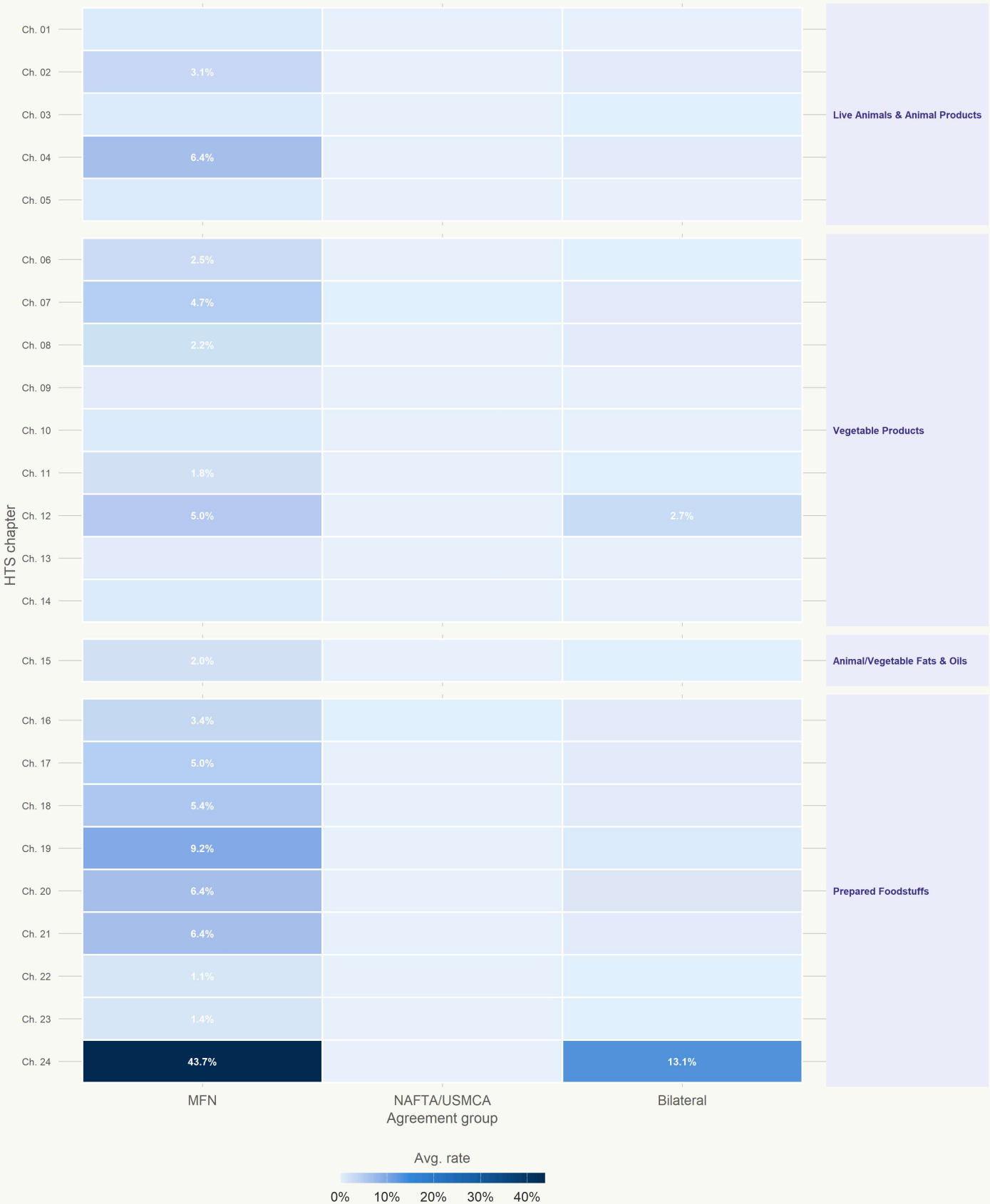
5.3 Agreement coverage heatmap

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Contents

Tariff rates by HTS chapter and agreement group

Each cell = mean ad valorem rate for that chapter–agreement pair.



Source: USITC · TidyTuesday 2026 W17 · #TidyTuesday

Figure 6: Average ad valorem rate by HTS chapter and agreement group. Darker = higher rates.

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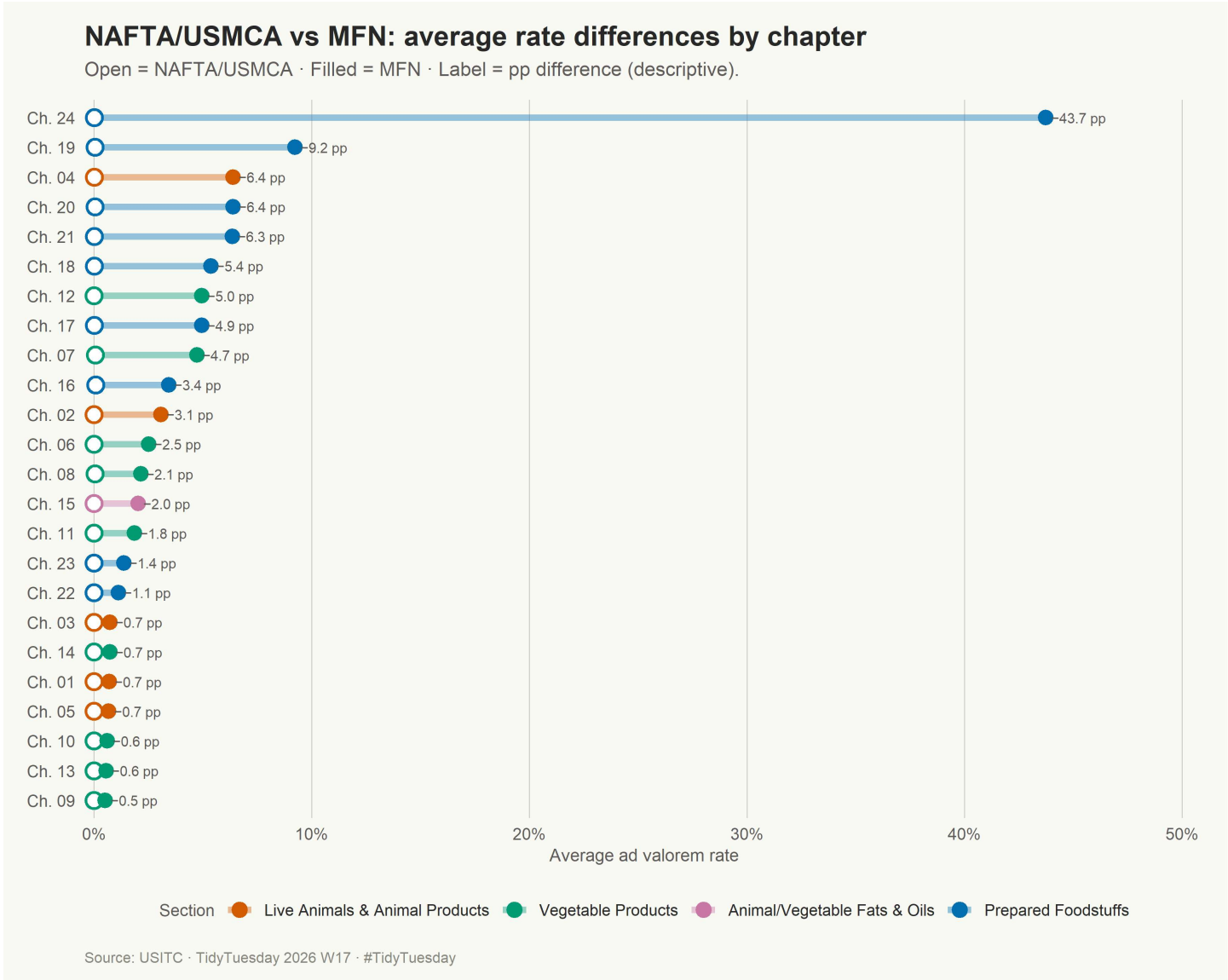


Figure 7: MFN (filled) vs NAFTA/USMCA (open) average rates per chapter. Segment = pp difference.

5.5 Tariff volatility – top 30 lines

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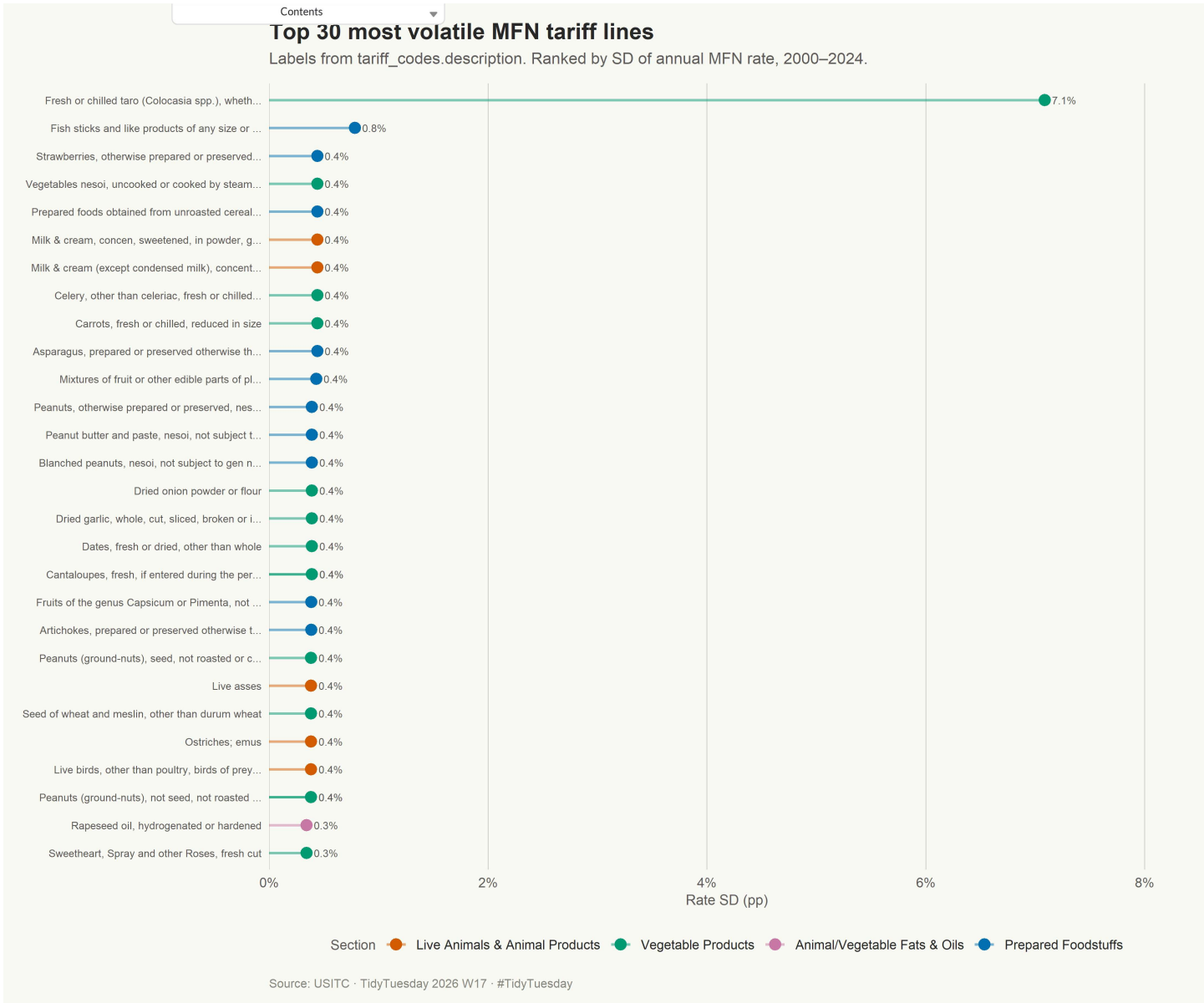


Figure 8: Top 30 MFN tariff lines by rate SD. Labels use product_desc from tariff_codes.csv.

5.6 NAFTA → USMCA transition

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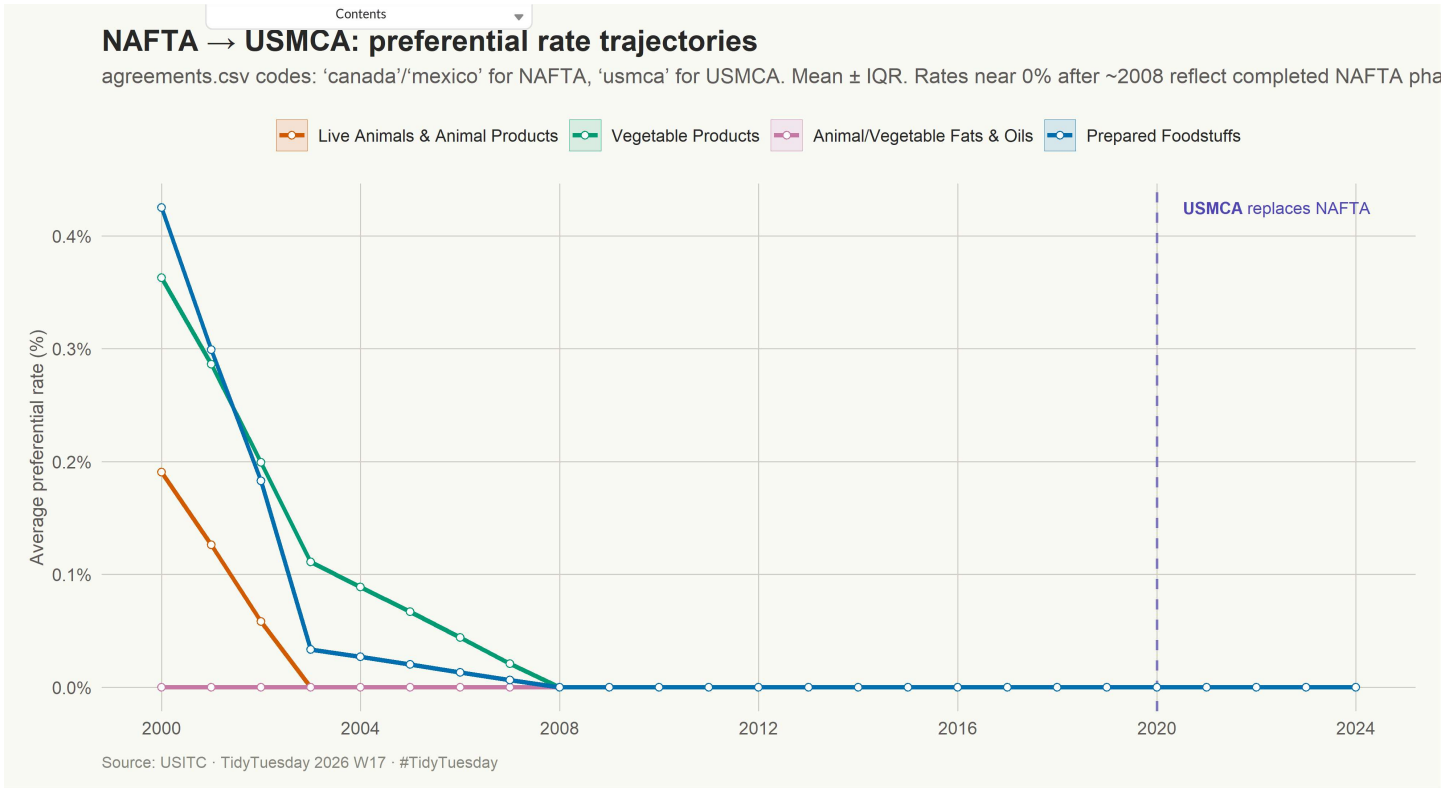


Figure 9: NAFTA/USMCA preferential rate trajectories. Dashed = USMCA effective date (2020).

5.7 Biggest rate movers

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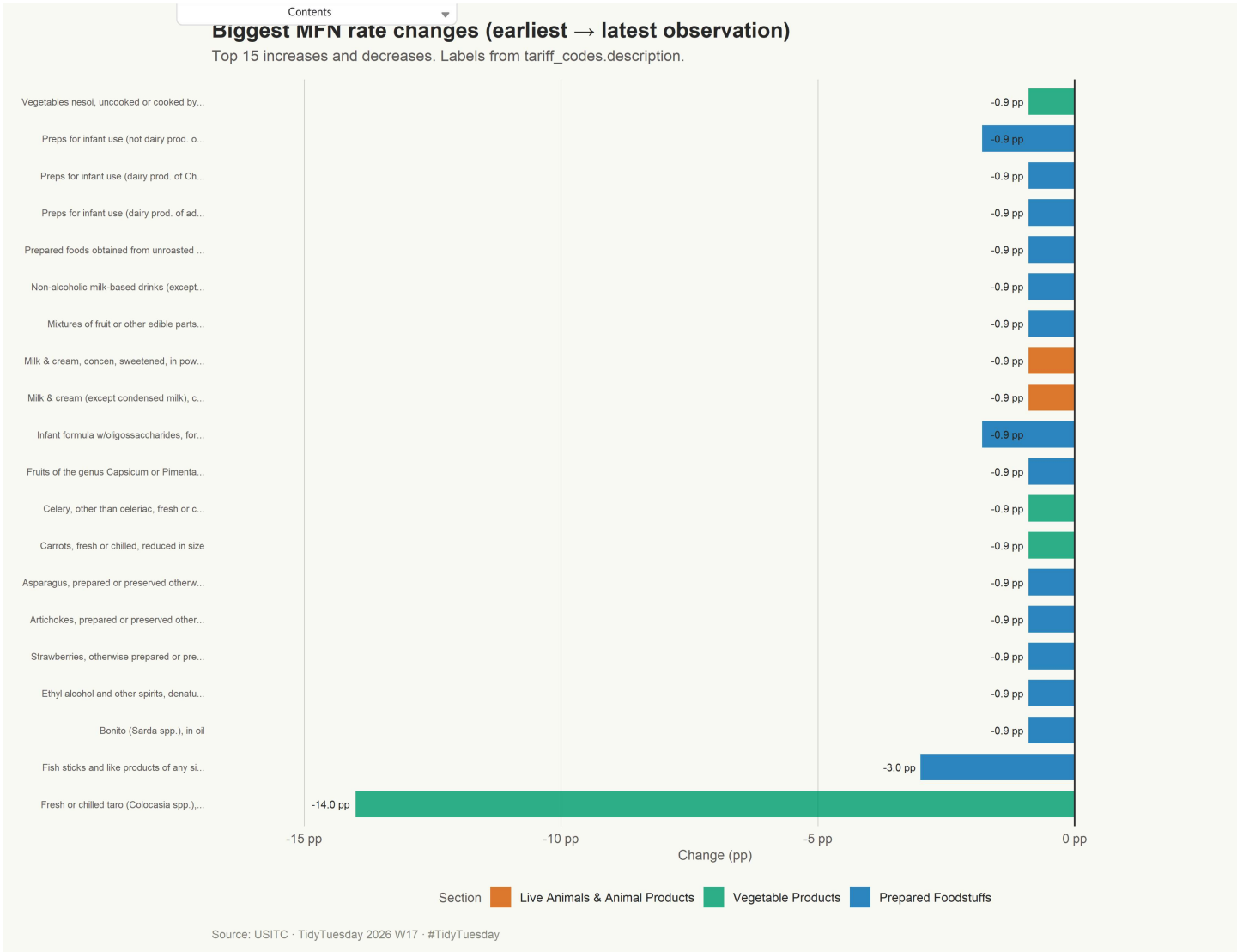


Figure 10: Top 15 MFN rate changes. Labels from tariff_codes.description via product_desc.

5.8 Agreements vs MFN rate — bubble

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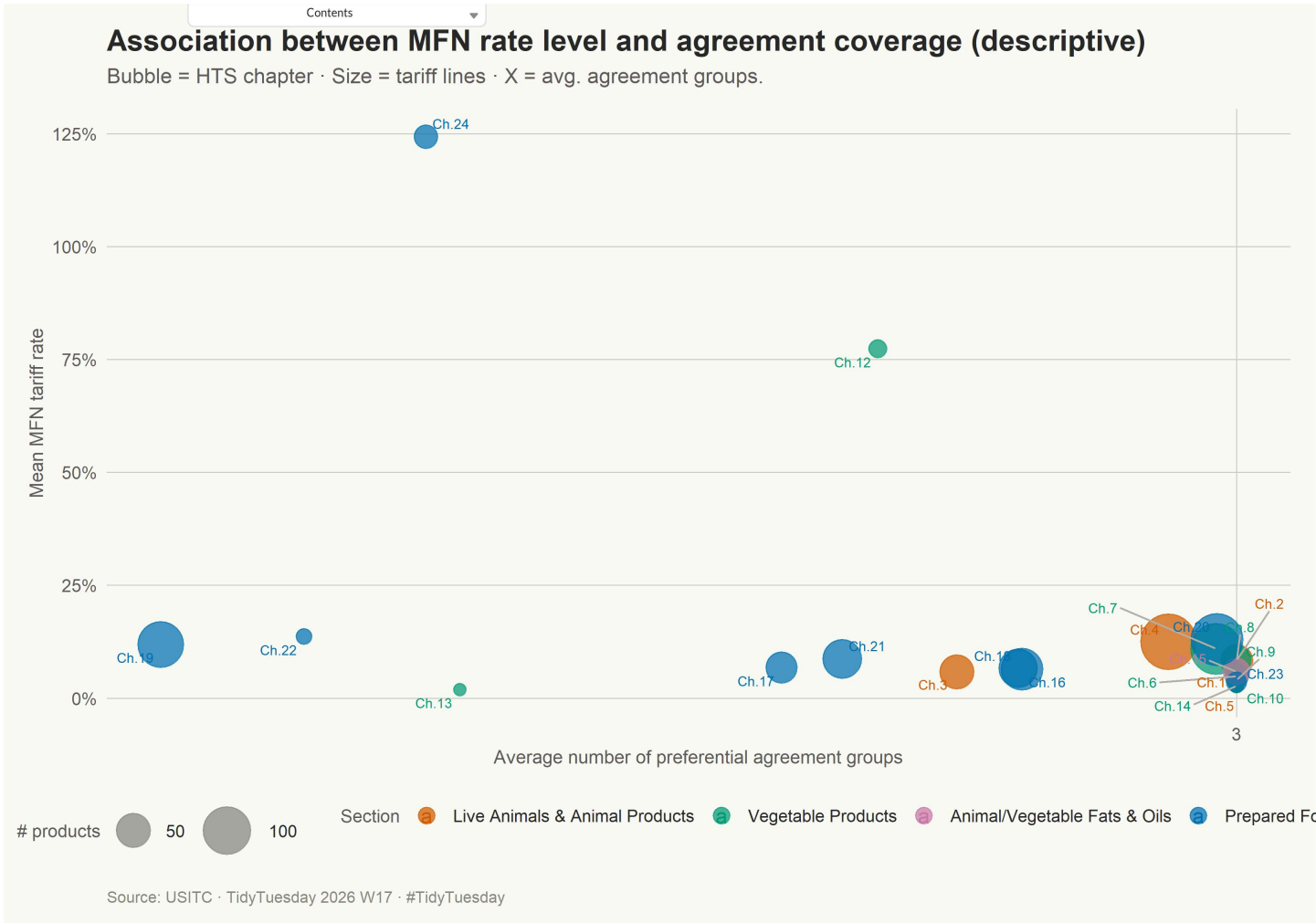


Figure 11: Association between MFN rate level and preferential agreement group coverage — descriptive only.

5.9 Tariff type composition

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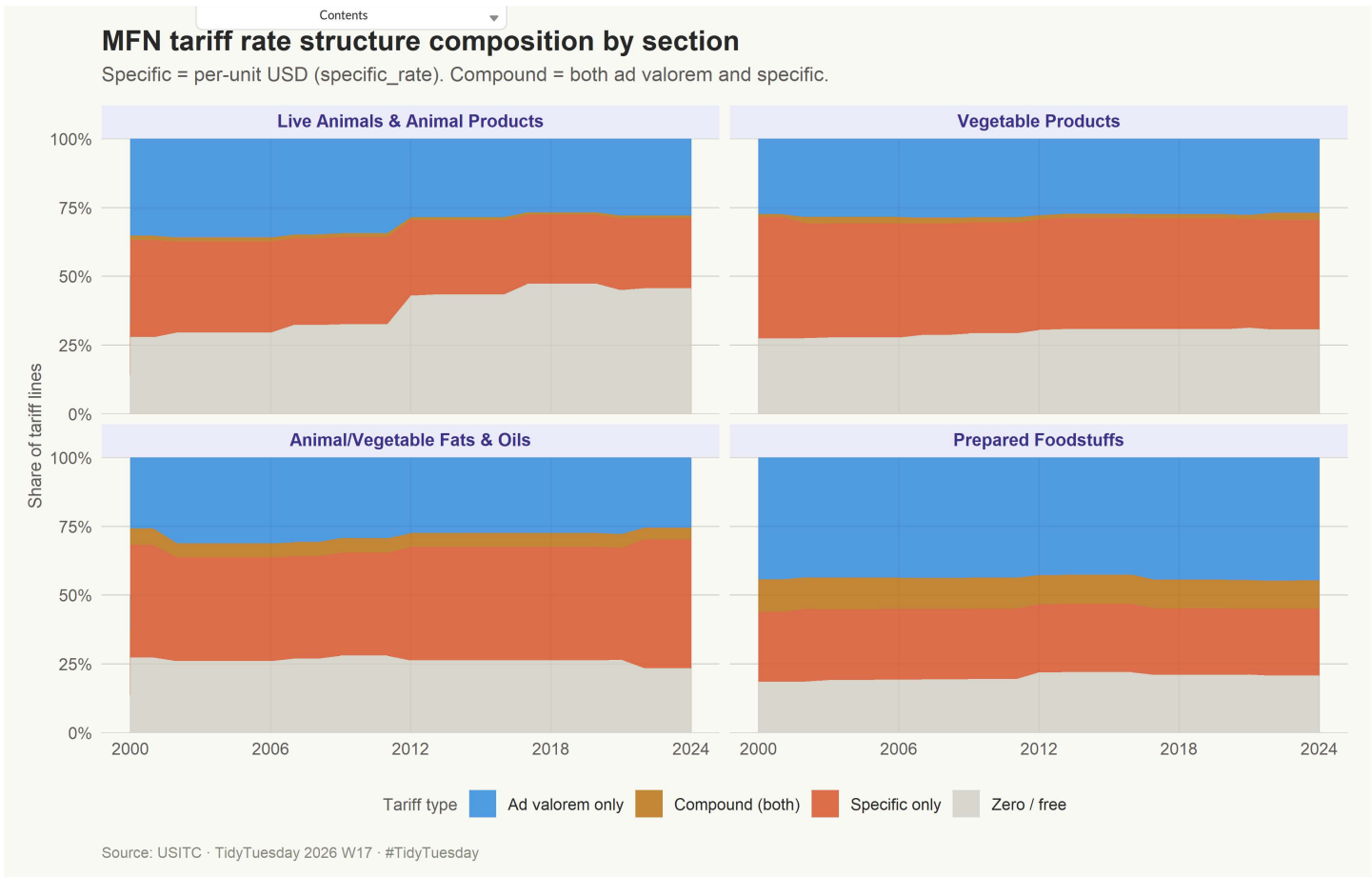


Figure 12: Share of MFN lines by rate structure. Specific = \$/unit from specific_rate column.

5.10 Rate distribution ridgelines

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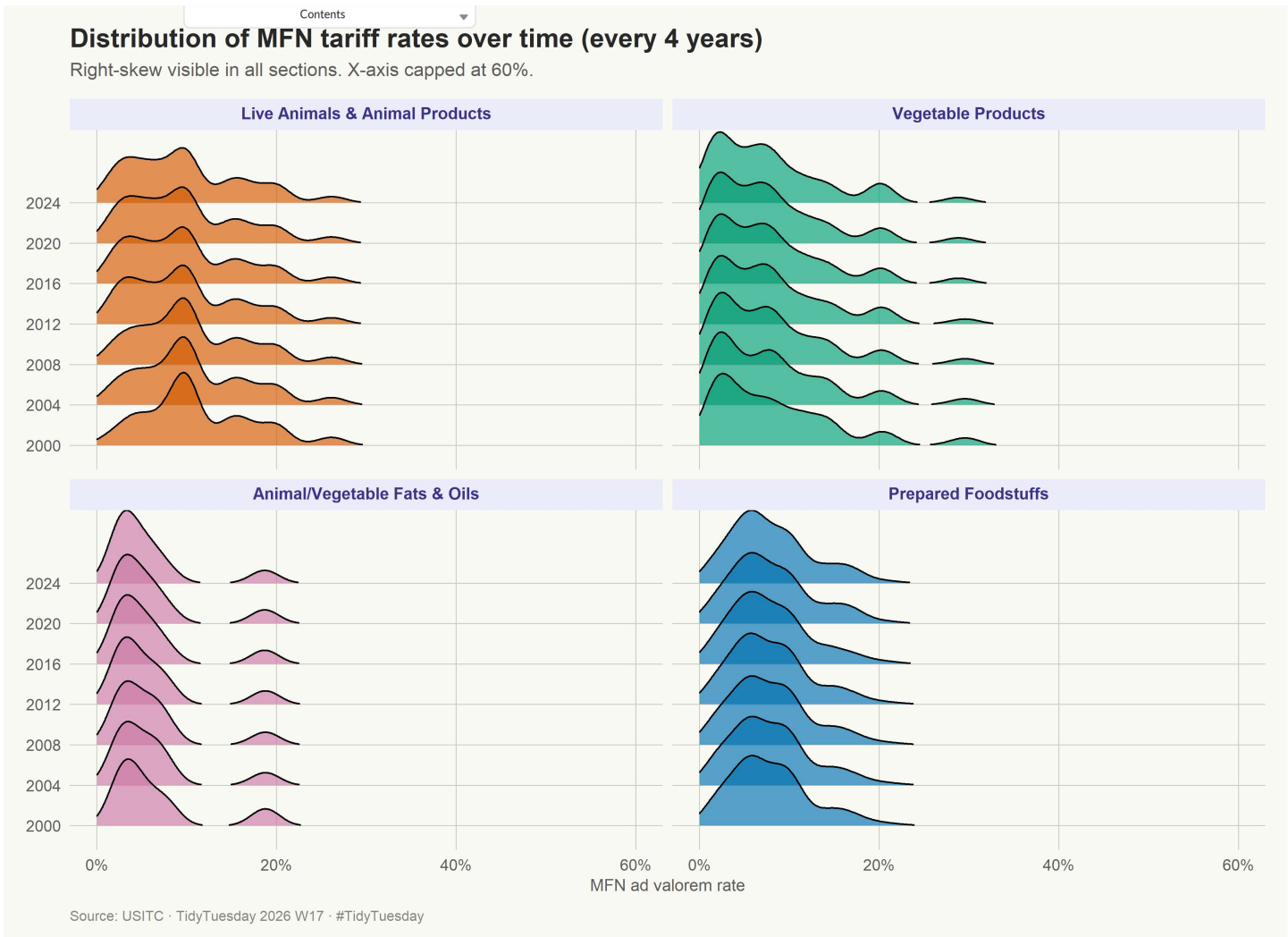


Figure 13: MFN ad valorem rate distributions every four years. X-axis capped at 60%.

6 Statistical Tests

Statistical significance does not imply practical importance or causation. Tests below assess consistency with random variation only.

6.1 Kruskal-Wallis

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Table 8: Kruskal-Wallis test: do MFN rate distributions differ across HTS sections?

Kruskal-Wallis test				
<i>H₀: all sections share the same rate distribution</i>				
Test	Statistic	df	p-value	Conclusion
Kruskal-Wallis rank-sum	2773.24	3	0.000e+00	Rate distributions differ across sections (p < 0.05)

6.2 Dunn post-hoc

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Table 9: Dunn pairwise comparisons (Bonferroni correction)

Dunn post-hoc (Bonferroni)

Pairwise section comparisons

Comparison	Z	p-adjusted	Significant
Animal/Vegetable Fats & Oils - Live Animals & Animal Products	-3.871	3e-04	Yes ✓
Animal/Vegetable Fats & Oils - Prepared Foodstuffs	-21.280	0e+00	Yes ✓
Live Animals & Animal Products - Prepared Foodstuffs	-39.228	0e+00	Yes ✓
Animal/Vegetable Fats & Oils - Vegetable Products	0.407	1e+00	No
Live Animals & Animal Products - Vegetable Products	9.210	0e+00	Yes ✓
Prepared Foodstuffs - Vegetable Products	47.961	0e+00	Yes ✓

6.3 Mann-Kendall trend test

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Table 10: Mann-Kendall monotonic trend test by section (2000–2024)

Mann-Kendall trend test (2000–2024)			
<i>H₀: no monotonic trend in annual mean MFN rate</i>			
Section	Kendall τ	p-value	Conclusion
Live Animals & Animal Products	−0.8002	0.0000	Significant decrease ↓
Vegetable Products	−0.2662	0.0737	No monotonic trend detected
Animal/Vegetable Fats & Oils	−0.7646	0.0000	Significant decrease ↓
Prepared Foodstuffs	−0.7709	0.0000	Significant decrease ↓

7 Interactive Exploration

7.1 Linked chapter explorer

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Filter by section:

► Show / hide code

Table 11: MFN rate statistics by HTS chapter — select rows to highlight in scatter below

CopyCSV

Search:

	Chapter	Section	Products	Mean_rate	Median	SD	Max	
1		Live Animals & Animal Products	59		0.69		0	1.646.8
2		Live Animals & Animal Products	125		3.08		0	6.3526.4
3		Live Animals & Animal Products	405		0.73		0	2.2615
4		Live Animals & Animal Products	271		6.39		4.2	7.1425
5		Live Animals & Animal Products	23		0.67		0	1.335.1
6		Vegetable Products	40		2.52		1.4	2.697
7		Vegetable Products	234		4.74		0	6.7629.8
8		Vegetable Products	164		2.16		0	5.4329.8
9		Vegetable Products	81		0.51		0	1.466.4
10		Vegetable Products	40		0.6		0	2.0711.2
11		Vegetable Products	50		1.85		0	3.3612.8
12		Vegetable Products	111		4.96		0	25.91163.8

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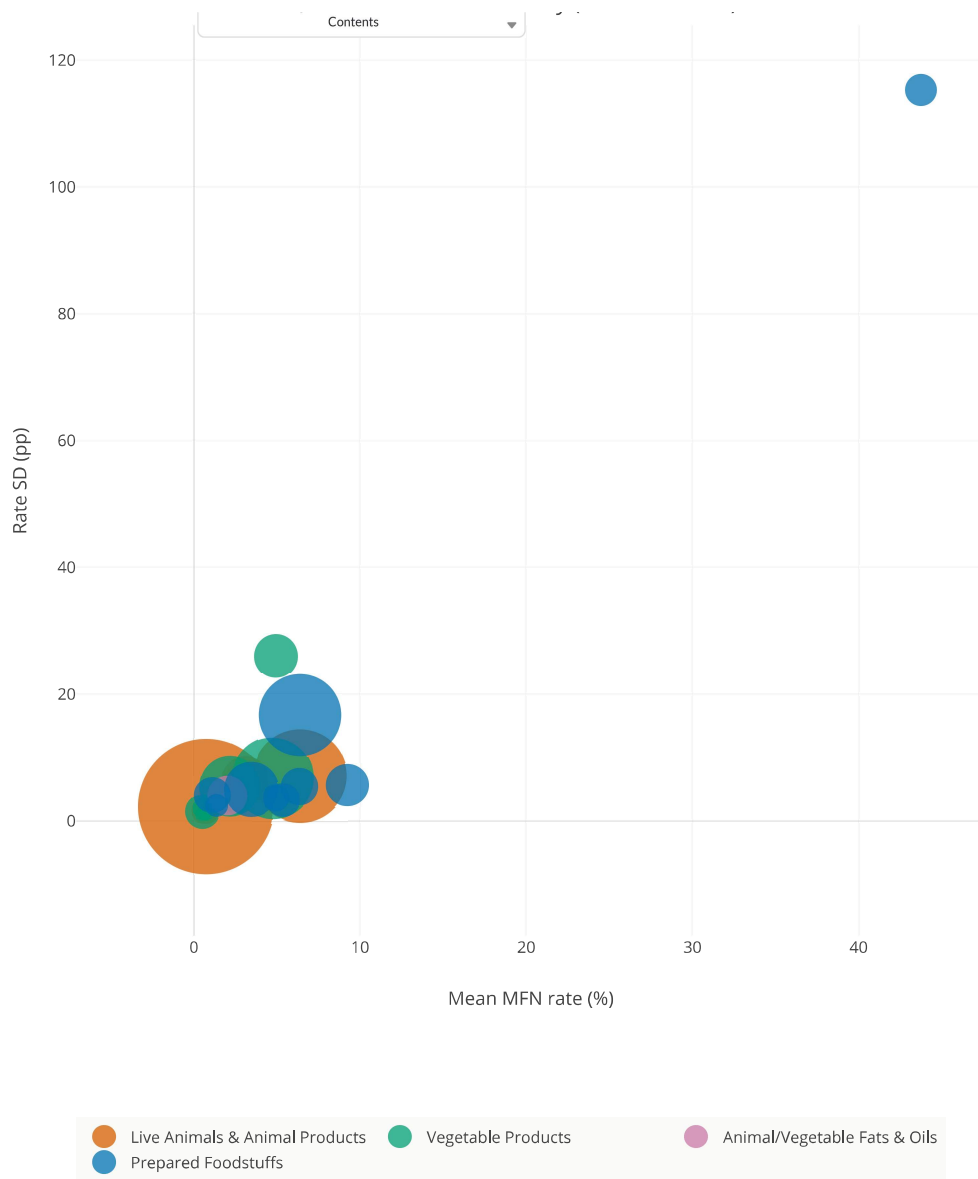
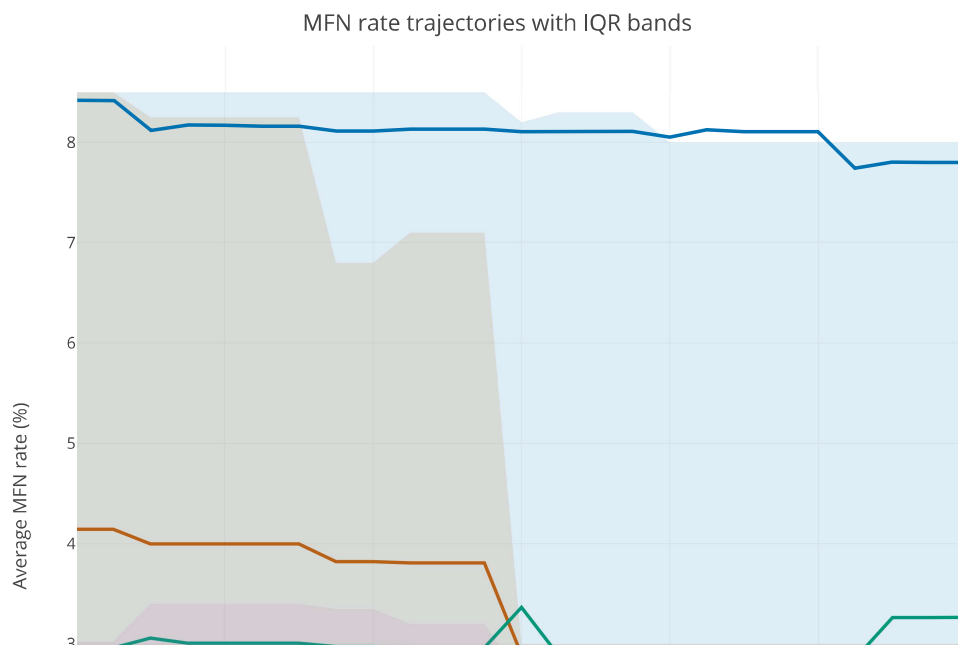


Figure 14: Linked scatter — select table rows to highlight chapters.

7.2 Interactive trajectories with IQR

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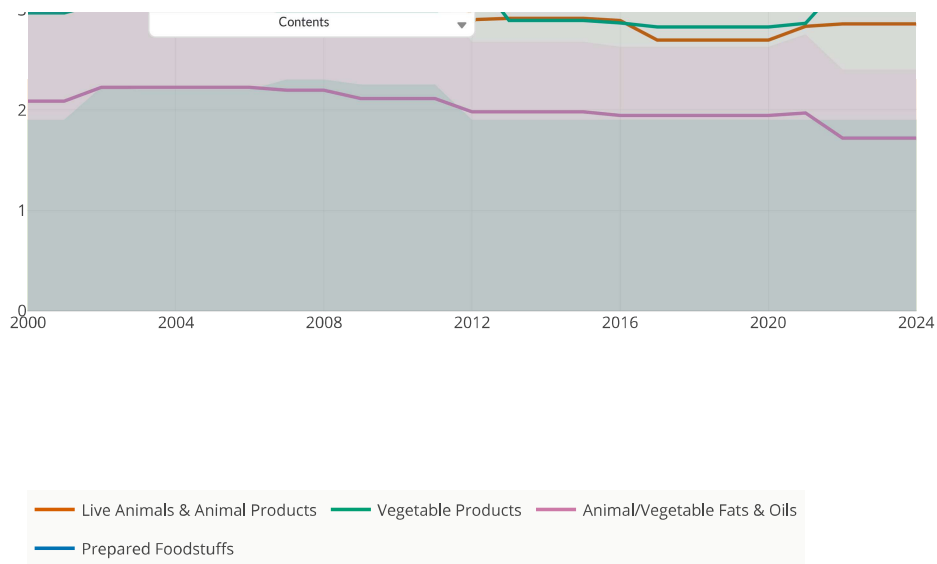


Figure 15: Interactive MFN trajectories with IQR bands — unified hover tooltip.

8 Composite Dashboard

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US Agricultural Tariffs — #TidyTuesday 2026 Week 17

USITC Tariff Database · HTS Ch. 1–24 · Descriptive analysis — no causal claims.

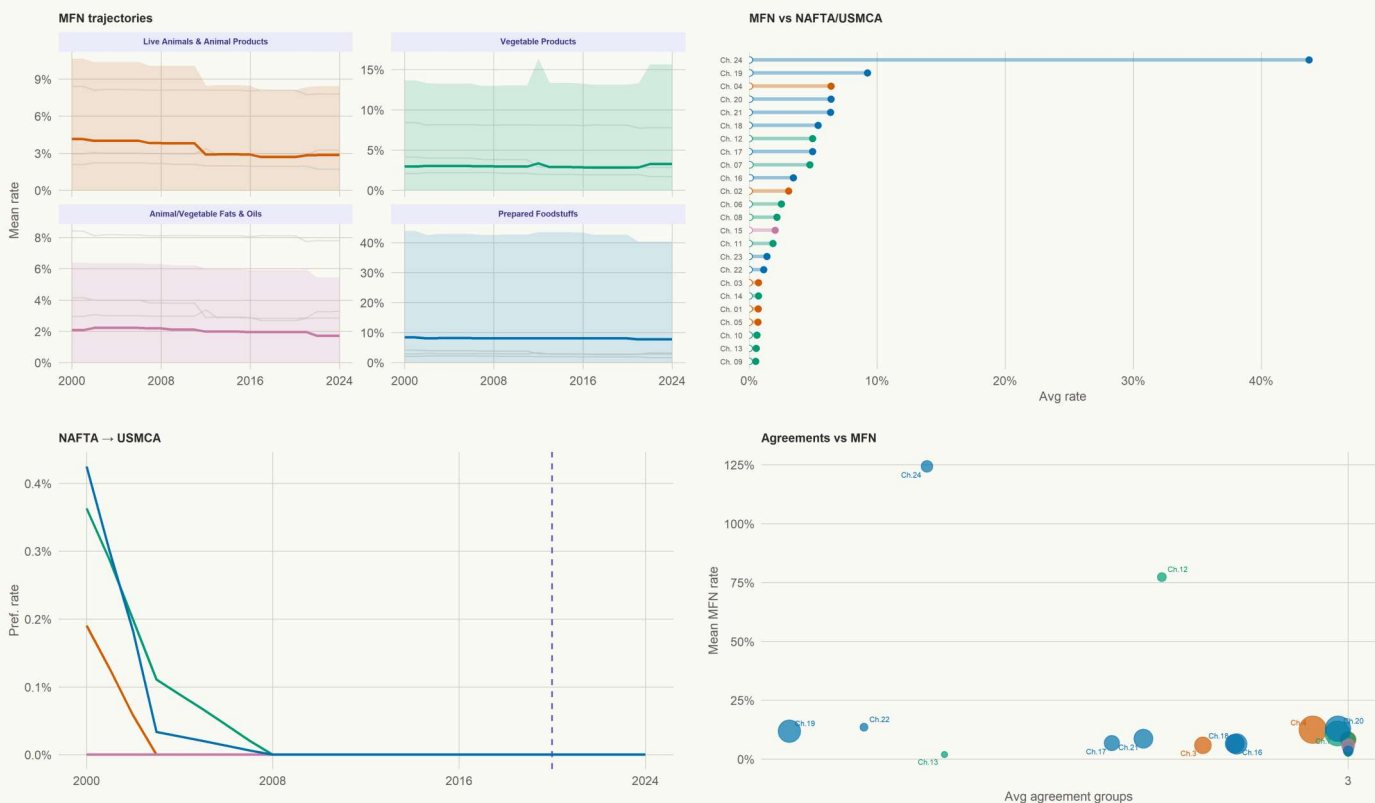


Figure 16: Four-panel summary dashboard — #TidyTuesday 2026 Week 17.

9 Key Findings

All seven points are **descriptive findings** from tidying and plotting the data as recommended by the TidyTuesday guidelines. No causal claims are made. Moderating variables not in this dataset — trade volumes, exchange rates, WTO negotiations, commodity market cycles — affect the patterns observed.

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Table 12: Seven descriptive findings — #TidyTuesday 2026 Week 17

Descriptive Findings — #TidyTuesday 2026 W17	
Data tidying and plotting practice — no causal claims	
# Finding	Technique practised
1 Live Animals & Animal Products show persistently higher average MFN rates across all years	Data tidying + line/ribbon charts
2 NAFTA/USMCA agreement codes are associated with lower average rates than MFN across most chapters	Pivoting + dumbbell charts
3 Prepared Foodstuffs show the least year-to-year rate variability in the dataset	Summarisation + heatmaps
4 Rate volatility (SD over time) is concentrated in ~30 tariff lines out of thousands	Ranking + lollipop charts
5 The 2020 NAFTA → USMCA transition coincides with no discontinuous jump in the rate series	Date handling + annotated time series
6 Chapters with higher MFN rates tend to have more preferential agreement groups (descriptive association)	Joining datasets + bubble charts
7 MFN rate distributions are statistically distinguishable across sections (Kruskal-Wallis, Dunn tests)	Statistical testing with nonparametric methods

10 How to Share This Work

🔥 Sharing your #TidyTuesday work

Following the [TidyTuesday sharing guidelines](#):

1. Post your output with [#TidyTuesday](#) on social media
2. Tag [@R4DSCommunity](#) and [@eyegonk](#) (Twitter/X or Mastodon)
3. Share your source code on GitHub
4. Link to the dataset: <https://github.com/rfordatascience/tidyuesday/blob/main/data/2026/2026-04-28/>

Suggested post text: > 🇺🇸 My #TidyTuesday Week 17 2026 submission by @eyegonk — exploring US agricultural tariffs from the USITC database. MFN trajectories, NAFTA→USMCA transitions, Lorenz curves, statistical tests across HTS Ch. 1–24. Code: [your link] 📊 #rstats #dataviz

Dataset citation: > USITC Tariff Database via TidyTuesday 2026, Week 17. > <https://github.com/rfordatascience/tidyuesday/blob/main/data/2026/2026-04-28/readme.md>

Rendered with: [Quarto](#) · R 4.6.0 · ggplot2 · plotly · gt · crosstalk · patchwork

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11 Reproducibility

Session info

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